

SHM – A Short Introduction

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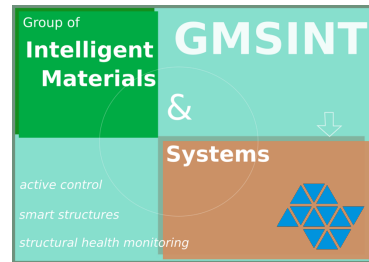
About me



Bachelor
2012 - 2016



PhD
2017 - 2021



Adv. Samuel da Silva



Post-Doc
2021 – 2022



Associate Professor
Since 2022



Research Internship
2019 - 2020

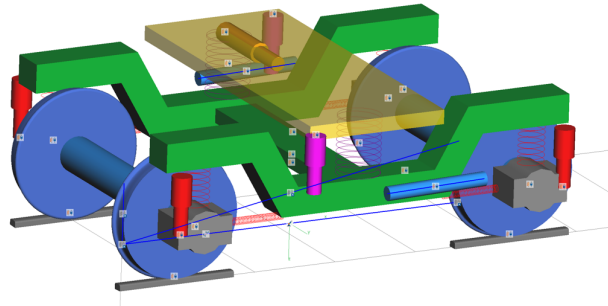


Adv. Gaël Chevallier

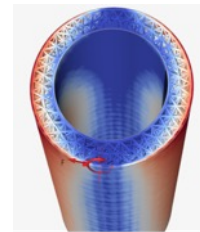
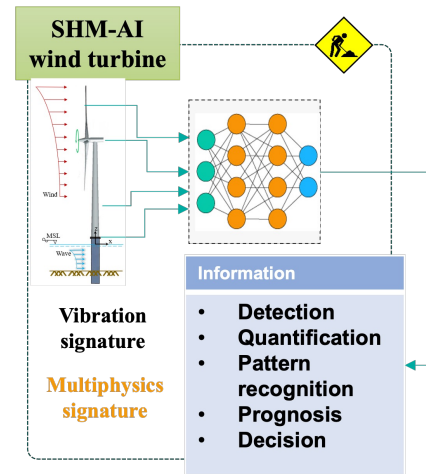
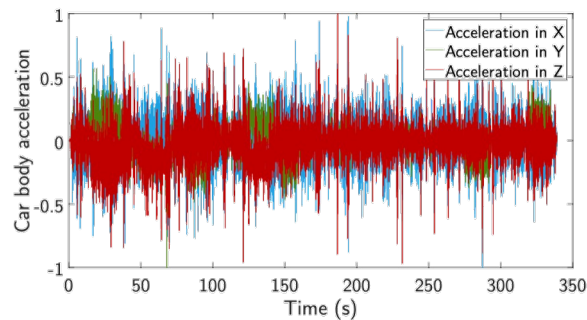
From Virtual Homologation to Digital Twins

VIRTUAL HOMOLOGATION

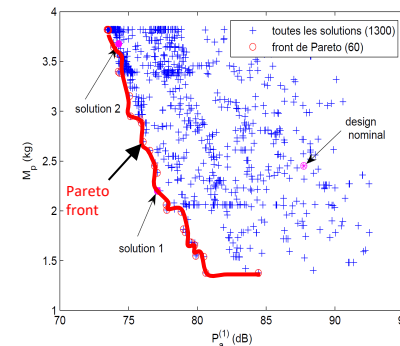
For railway dynamics



ALSTOM



SAFRAN

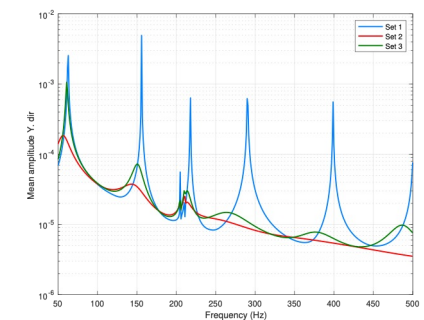
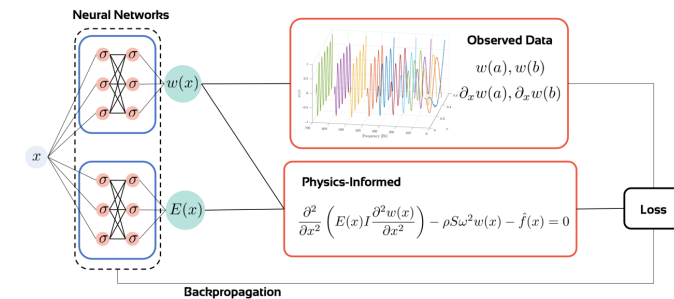


DATA-DRIVEN MODELING

Structures, acoustics, multiple scales modelling, anomaly detection

PHYSICS INFORMED MACHINE LEARNING

Missing physics, hybrid modeling, contact problems



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SHM – A Short Introduction

Outline

- 1 Motivation: why monitor structures?
- 2 Structural Health Monitoring: concepts and levels
- 3 The SHM process: sensing, data and decisions
- 4 SHM methods: vibration, guided waves, impedance
- 5 Case study: wind turbine blades
- 6 Challenges and outlook

Motivation: why monitor structures?

Society depends on an extensive infrastructure system

- Bridges, highways, buildings and dams
- Aircraft, offshore platforms and wind turbines

Structural failure leads to

- Casualties
- Direct and indirect economic losses
- Threats to society's well-being on a massive scale

Design guarantees mechanical resistance, but structures **deteriorate**: corrosion, fatigue, impact...

Images: Adapted from S. da Silva, Introduction to Structural Health Monitoring, UNESP (2023)



Damage: definition and types

Damage = changes in

- Material properties
- Geometric properties
- Boundary conditions
- System connectivity

If not repaired, damage can evolve into **failure**

Image: Adapted from S. da Silva, Introduction to Structural Health Monitoring, UNESP



Discrete events

- Earthquakes, hurricanes and storms
- Fire
- Impact
- SHM helps assess the prognosis of the structure after the event

Continuous evolution

- Operational conditions (fatigue, corrosion)
- Incorrect use
- Design or manufacturing errors
- Deterioration from normal use

SHM: Current Damage measurement

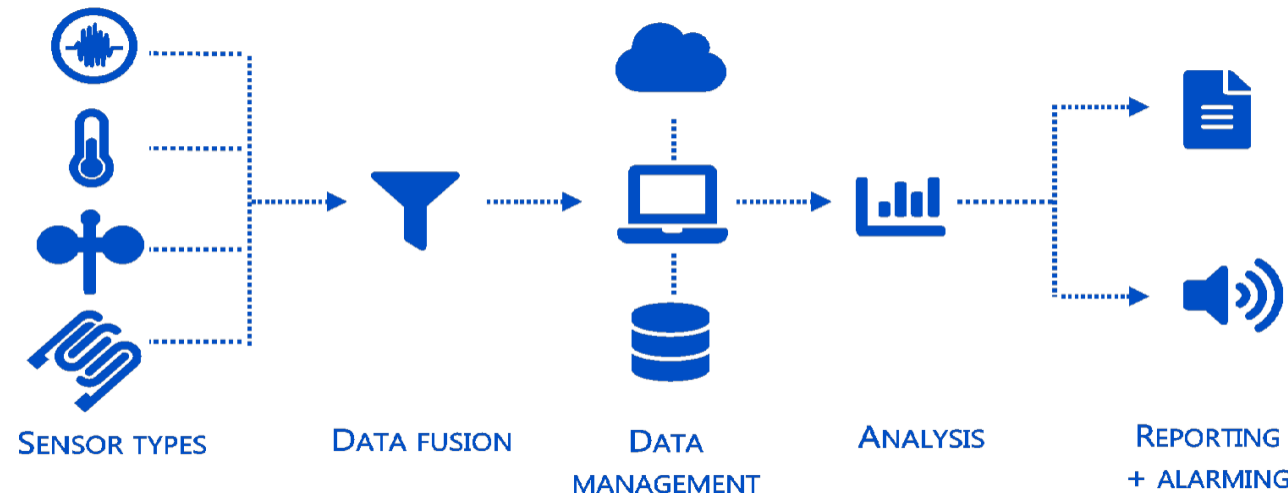
Visual inspection
Localized experimental

- Ultrasonic
- Acoustic Emission



What is SHM

- A method for evaluating and monitoring structural health, aimed at early damage detection
- Widely applied across engineering sectors: responds to adverse structural changes, improving reliability and life-cycle management
- Non-destructive, in-situ structural evaluation
- Uses several sensors, embedded in or attached to the structure



Body Analogy/ SHM



- Monitors the patient's health
- Uses medical equipment to check overall health
- Prescribes corrective medicine if required

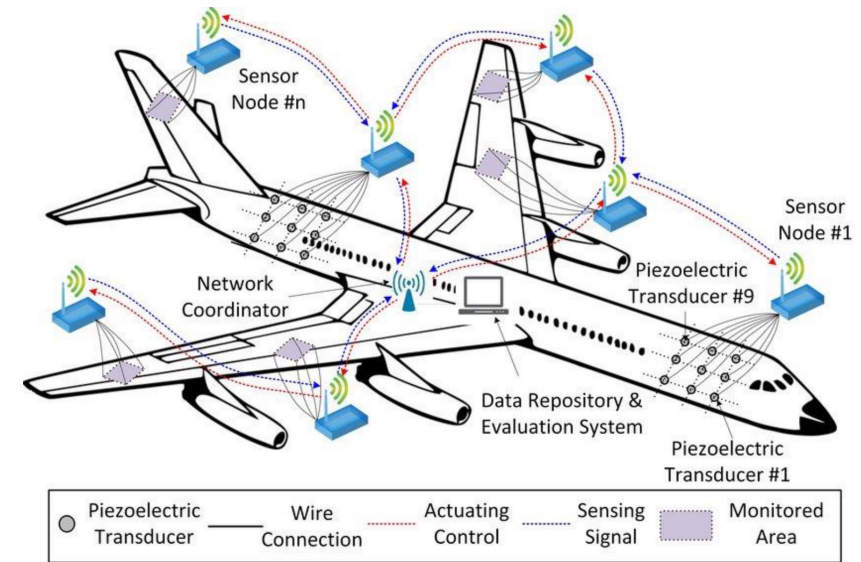


- Monitors the condition of structures
- Uses sensors to check overall structural health
- Corrects the situation in case of excessive stress or deformation

Advantages and applications of SHM

Advantages

- Early damage detection
- Better understanding of in-situ structural behaviour
- Less downtime for inspection and repair
- Assurance of structural strength and serviceability
- Rational maintenance and management strategies
- More effective allocation of scarce resources
- Encourages the use of new and innovative materials



Applications

- Aerospace: airframes, engines, composites
- Civil: bridges, dams, skyscrapers
- Mechanical systems and machinery
- Energy: wind turbine blades

Levels of SHM: from detection to prognosis

Level I · Detection Is there any change?

Level II · Localization Where is it?

Level III · Classification What type of damage?

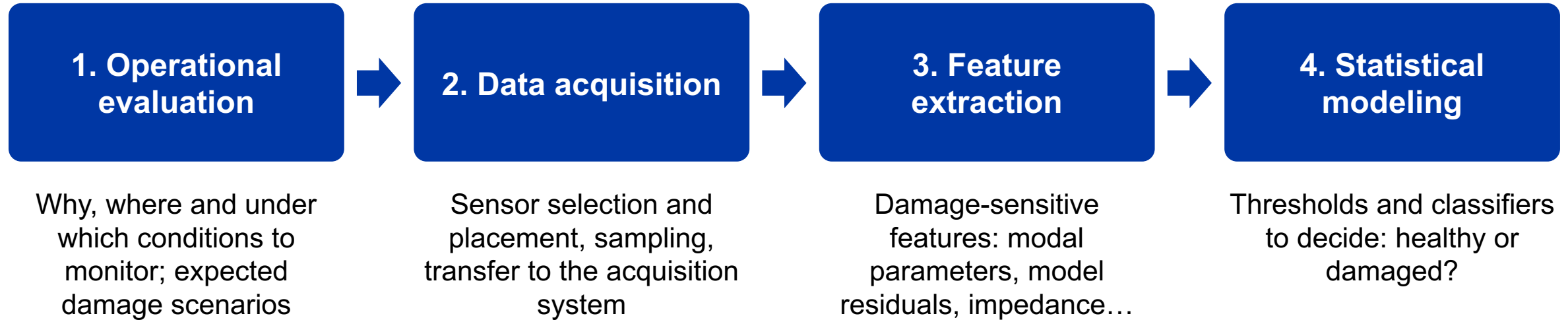
Level IV · Quantification How severe is it?

Level V · Prognosis What is the remaining useful life?

Higher levels require a mathematical model: white-box (physics-based), black-box (data-driven) or gray-box (hybrid)



The SHM process

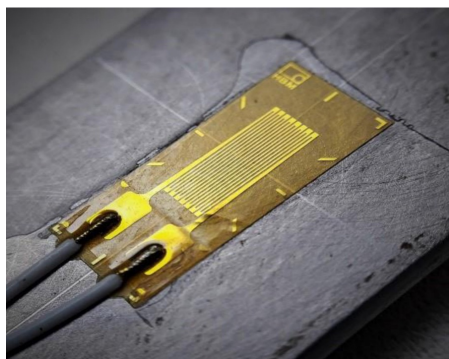


Practical challenges

Complex (anisotropic) materials · Nonlinear behaviour · Uncertainties and operational variability

Data acquisition: what is monitored and how?

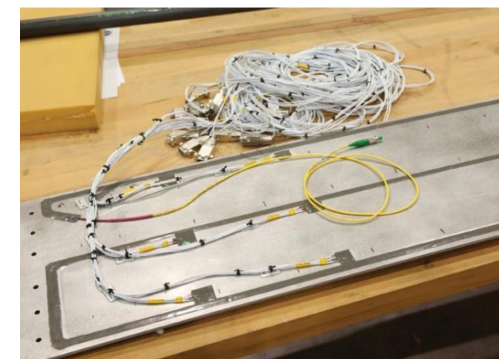
Quantity	Why monitor it	Typical sensor
Load	Magnitude and configuration of applied forces	Load cells, or inferred from strain
Strain / deformation	Intensity of deformation; excessive or unexpected deformation	Strain gauges, fiber optic sensors
Acceleration	Dynamic response of the structure to loads	Accelerometers
Temperature	Causes deformation and affects strain readings	Thermocouples, thermistors
Acoustic emission	Locates cracking noise by triangulation	Acoustic emission sensors
Wind speed / pressure	Governs long-span bridges and tall buildings	Anemometers



Strain gauges



Accelerometer



Fiber optic sensors

Monitoring strategy and data communication

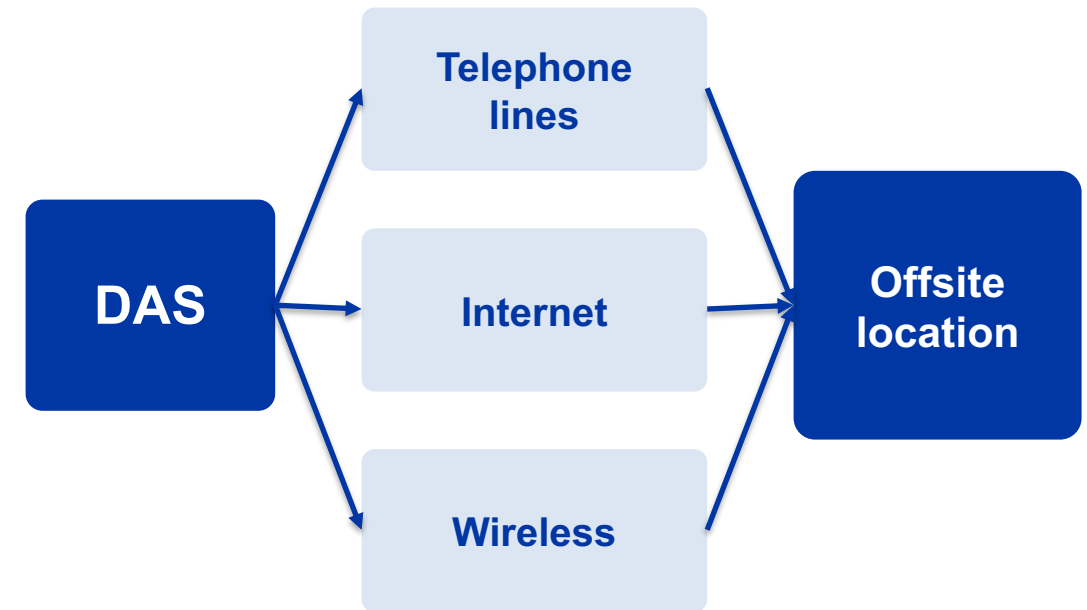
Periodic monitoring

- Tracks detrimental changes at set intervals (days, weeks, months, years)
- Lower cost; changes between visits can be missed

Continuous monitoring

- Ongoing over an extended period
- For critical structures or when integrity is in doubt
- Higher cost and complexity
- Data stored on site or sent for real-time analysis

Data are transferred from the data acquisition system (DAS) to an offsite location, allowing remote monitoring without site visits.



SHM methods: an overview

Vibration-based

Global method

Natural frequencies, mode shapes, time-series models

Guided waves

Local method

Ultrasonic waves sent and received by a PZT network

Electromechanical impedance

Local method

PZT acts as actuator and sensor at the same point

All these approaches involve:



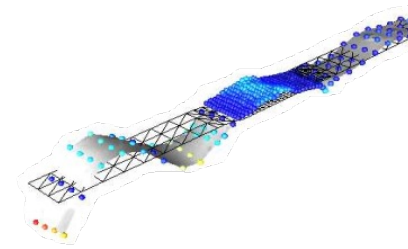
Training strategies: supervised (labels known) · unsupervised (labels unknown) · semi-supervised

Vibration-based SHM

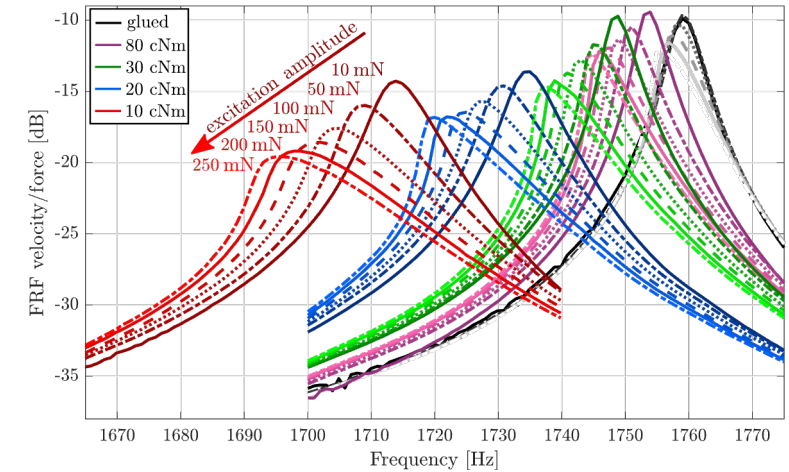
Sensors placed strategically measure the dynamic response (displacement, velocity, acceleration, strain); changes in that response reveal damage.

Advantages

- Most popular: inexpensive sensors
- Features easy to extract (modal parameters)
- Well correlated with finite element models

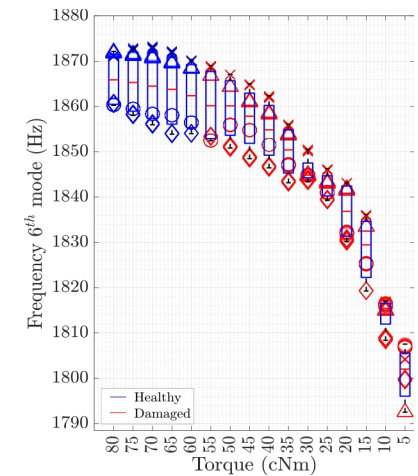
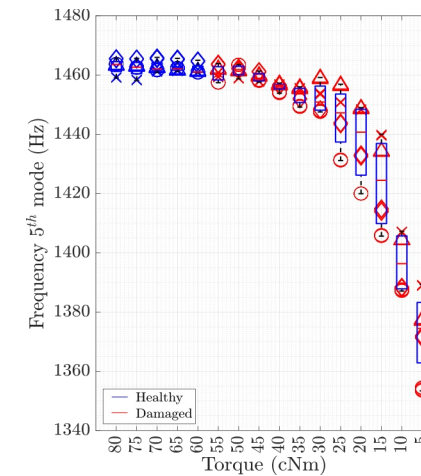


6th: 1758 Hz



Limitations

- Global method: less sensitive to small defects
- Sensitive to environmental effects
- Nonlinearities can appear



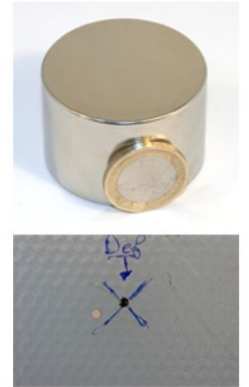
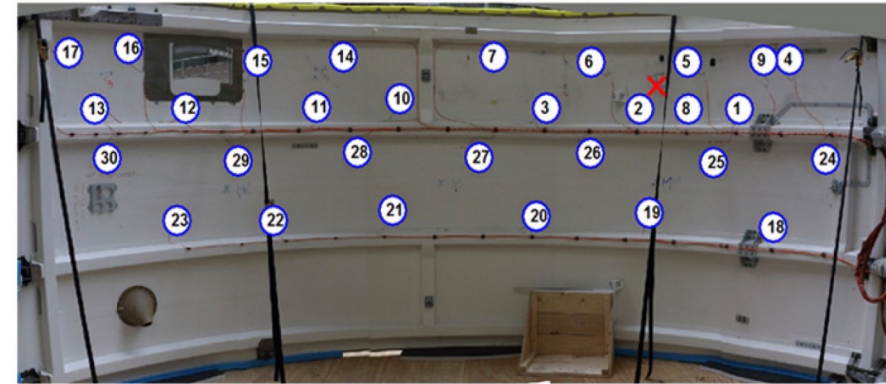
Guided-wave SHM

Advantages

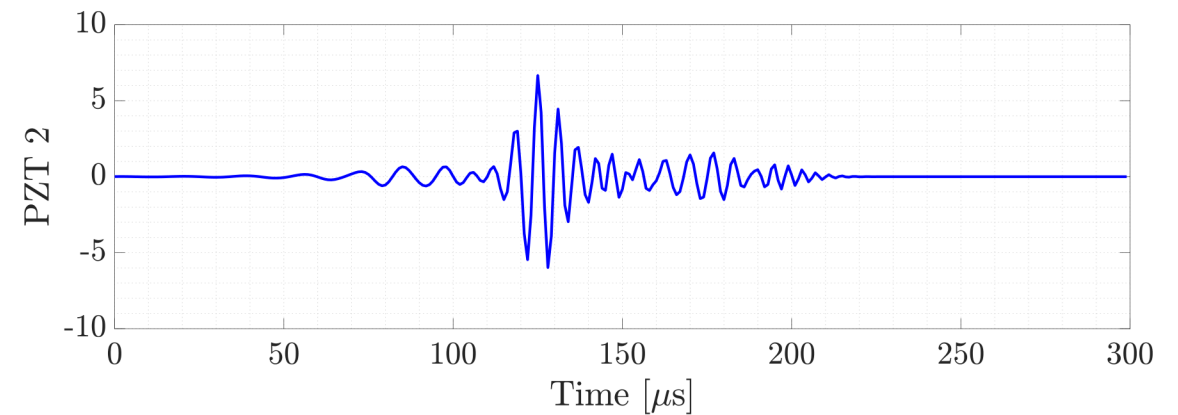
- Local method: sensitive to very small defects
- Features extracted directly from the signals
- Can cover large areas with a PZT network

Limitations

- Requires a network of sensors
- Reflections at boundaries complicate signals
- Needs a controlled input (frequency and amplitude)



PZT network on a nacelle fan cowl



Tone-burst response measured at PZT 2

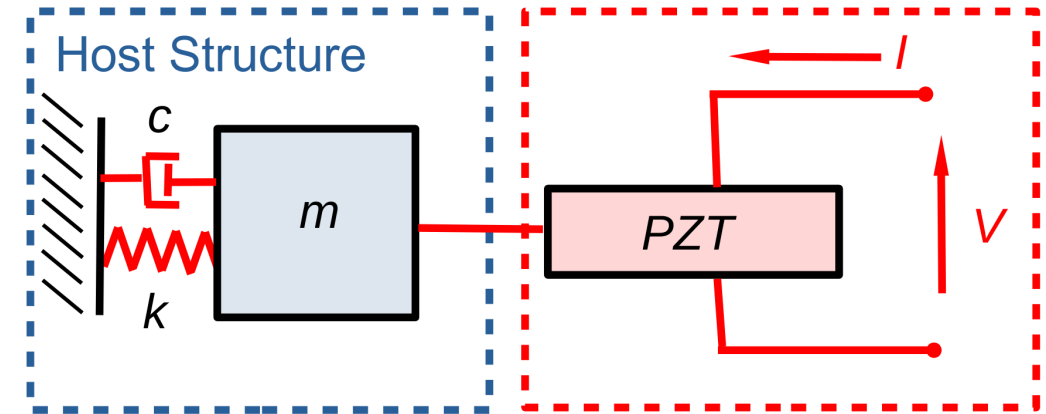
Electromechanical impedance-based SHM

Advantages

- The same PZT acts as actuator and sensor
- Features easy to extract from impedance signatures
- Many classifiers can be applied

Limitations

- Challenging to model
- Limited for quantification and prognosis
- Sensitive to temperature
- Requires post-processing



Aeronautical panel instrumented with 8 PZTs

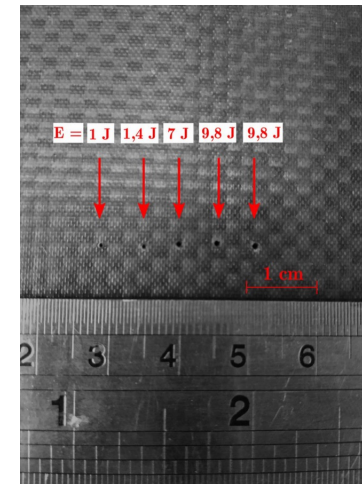
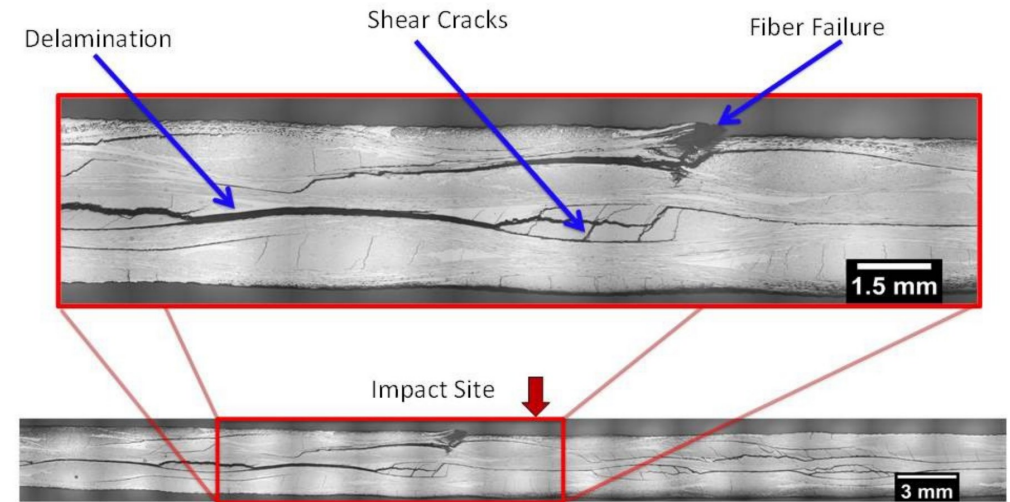
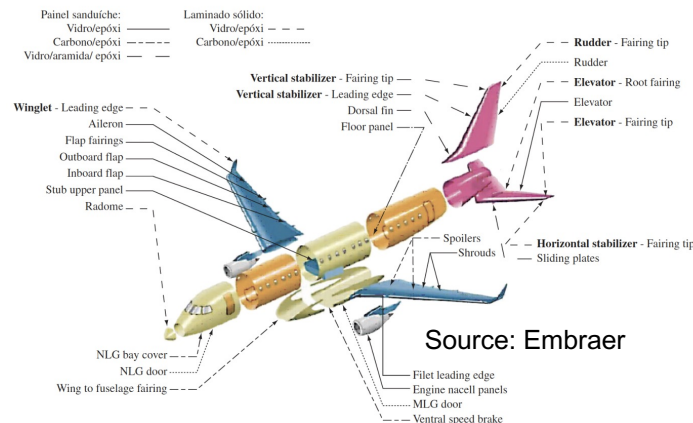
Composite structures: why they need SHM

Green structures: fuel efficiency + high stiffness-to-weight ratio

- Aircraft: Boeing 777, Airbus A380, Embraer 170
- Also wind energy, agriculture and automobiles

Typical defects

- Delamination, matrix cracking and broken fibers significantly reduce stiffness
- Can be caused by a low-velocity impact
- Usually internal and visually undetectable



Sources: blogs.nottingham.ac.uk/aerospace/ (2018); safran-nacelles.com; Adapted from S. da Silva, Introduction to Structural Health Monitoring, UNESP

Wind turbine blades: a critical application

Wind energy industry

- Larger rotors → more energy and lower cost per kWh

Blades made of composite materials

- Complex internal structure with multiple failure modes
- Hard to identify and predict failure early

Delamination, matrix cracking, fiber debonding... →
catastrophic failure



Blade failure in Piauí, Brazil (Feb. 2020)

Challenges and outlook

Open challenges

- Environmental and operational variability
- Nonlinearities and propagation of uncertainties
- Reaching quantification and prognosis levels

Complementary techniques

- Digital image correlation, X-ray and ultrasonic NDT

Trends

- Frugal AI
- Training with laboratory data + finite element models, testing on real structures
- Digital-twins





Thank you for your attention!

Questions?

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